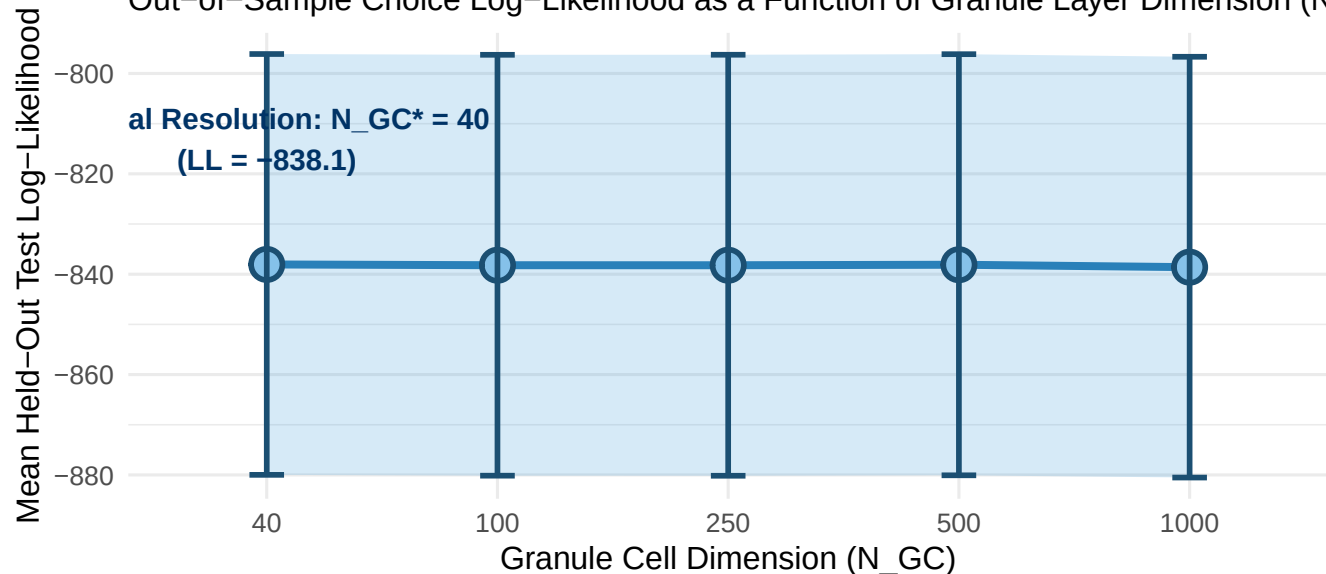


## A. Monte Carlo Dimensional Scaling Curve (K=10 Folds, Held-Out N=1000)

Out-of-Sample Choice Log-Likelihood as a Function of Granule Layer Dimension ( $N_{GC}$ )



## B. Topological Ablation at Optimal Dimension ( $N_{GC} = 40$ )

Severing Recurrent Golgi Loops ( $W_{fb}=0$ ,  $W_{inh}=0$ ) and Fading Memory ( $\tau=0$ )

